All About Databases

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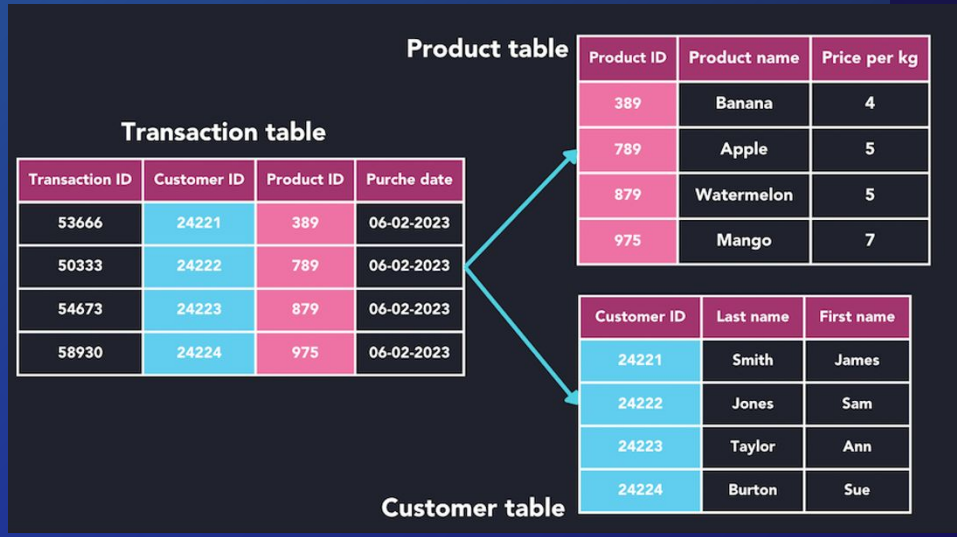
Relational Databases

- Proposed in 1970 by Edgar F.Codd at IBM
- Needed an efficient way to utilize disk space as storage was expensive
- Data is stored into rows and columns
- Tables connected by key that illustrate the relationship between them



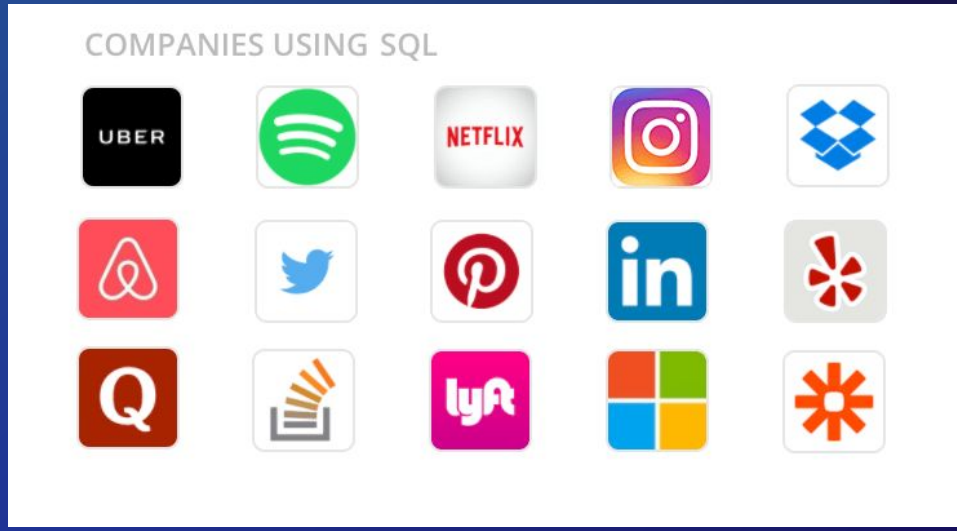
Transaction ID	Customer ID	Product ID	Purche date
53666	24221	389	06-02-2023
50333	24222	789	06-02-2023
54673	24223	879	06-02-2023
58930	24224	975	06-02-2023

Product ID	Product name	Price per kg
389	Banana	4
789	Apple	5
879	Watermelon	5
975	Mango	7

Customer ID	Last name	First name
24221	Smith	James
24222	Jones	Sam
24223	Taylor	Ann
24224	Burton	Sue

SQL

- Structured Query Language
- Invented in 1970s by Don Chamberlin and Ray Boyce at IBM
- Used to manage data in relational databases
- Allowed for an easy way to handle data manipulation, data definition, and access control



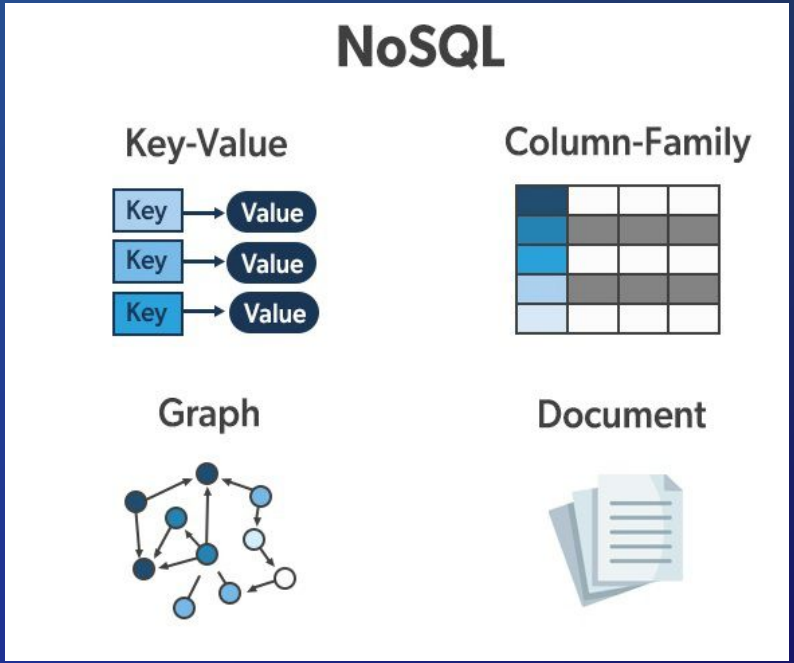
Popular SQL Databases

- Open-source relational database
- Widely used in web applications
- Uses structured tables with rows and columns
- Supports ACID transactions for data integrity
- Easy to learn, great for small-to-medium projects

- Relational database built for large businesses
- Industry standard in banking, finance, and healthcare sectors
- Known for high performance, scalability, and robust security
- Offers advanced tools for managing large amounts of data

Non -Relational Databases

- Popularized in the late 2000s by Web 2.0 companies to handle massive scale
- Needed an efficient way to scale horizontally across many servers
- Data is stored in flexible formats like documents, graphs, or key-value pairs
- Dynamic schemas store related data together instead of splitting it across tables

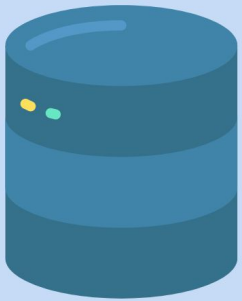


Popular NoSQL Databases

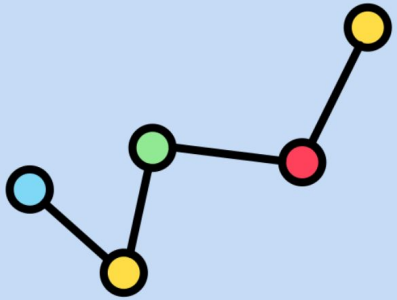
- Open-source document-oriented NoSQL database
- Stores data as flexible JSON-like documents (BSON)
- Schema-less design allows easy handling of unstructured data
- Popular in real-time apps, content management, and e-commerce

- In-memory key-value NoSQL data store
- Extremely fast, often used for caching and session management
- Supports data structures like strings, lists, sets, and hashes
- Used for instant, high-speed data access
- Examples - leaderboards, real-time analytics, message queues

Side by Side



SQL



NoSQL

Side by side

Feature	SQL Databases	NoSQL Databases
Flexibility (Scaling)	Vertical (Scaling up)	Horizontal (Scaling out)
Schema	Fixed / Rigid	Dynamic / Flexible (can manage various data types easily)
Data Management	Standard Query Language	Multiple methods (allows for creativity and flexibility)
Query Complexity	Generally simpler to write	Queries can be more complex to write
Database Properties	ACID	CAP Theorem
Ease of Use	Easy to learn and use	Steeper learning curve

Relational vs. NoSQL Use Cases

Relational	Non Relational
• Banking • Retail • ERP • CRM	• Web Scraping • Gaming • Social Media • Big data application • Real time analytics

Thanks!

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